Prateek Singh

Mechatronics Engineering

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Skills

- **Languages:** C • C++ • Verilog / VHDL • Java
- **Embedded & Tools:** STM32 • I2C • PWM • UART • AutoCAD • SolidWorks • Git

Projects

FPGA Night Vision Camera System | Capstone **Sept 2025 – Mar 2026**

Tang Primer 25K/20K FPGA, Verilog, OV5640 Camera, SON-LC02A Display

- Implemented a real-time camera pipeline in Verilog on a Tang Primer FPGA with an OV5640 image sensor
- Developed custom I2C master controller to configure the OV5640 camera for night mode operation
- Implemented DVP capture logic for YUV422 pixel streams synchronized to VSYNC/HREF/PCLK signals

8-bit Breadboard Computer **Sept 2024 – Aug 2025**

Discrete Logic CPU, ALU, 8-bit Registers, Clock module, RAM

- Built an operating Turing complete 8-bit computer with integrated circuits on breadboards
- Developed an assembly language that can be decoded by the 8-bit breadboard computer

6502 Breadboard Computer & VGA Graphics Card **Sept 2024 – Aug 2025**

6502 CPU, Assembly, Address Decoding, SRAM/ROM

- Built a functioning 8-bit computer using a 6502 microprocessor with discrete components
- Programmed the computer with assembly and executed programs stored in a breadboard built EEPROM
- Built a discrete logic video card capable of generating VGA signals using TTL logic components
- Implemented VGA timing generation using HSYNC, VSYNC, pixel clock for monitor output

Autonomous Mobile Robot Navigation System (MTE 544) **Sept 2025 – Dec 2025**

ROS2, Python, A Path Planning, Particle Filter Localization, PID Control, Rviz*

- Implementation an A* path planner and PID trajectory controller for autonomous robot navigation
- Built a particle filter localization system using LiDAR and odometry data to estimate robot position in real time

STM32 Quadcopter Firmware Drivers **Sept 2025 – Mar 2026**

STM32F4, C, Bare-Metal, I2C, PWM, USART, PID, MPU6050, LIS3MDL

- Programmed bare-metal STM32F4 drivers for I2C, PWM, and USART
- Implemented I2C communication for configuration of the MPU6050 IMU and LIS3MDL magnetometer
- Programmed PWM routines for motor control and USART interfaces for debugging

Experience

Mechanical Coordinator | Co-op | Toronto District School Board (TDSB) **May 2023 – Aug 2023**

- Coordinated HVAC system documentation across multiple school facilities

Lab Assistant | Co-op | Jenike & Johanson Ltd. **Sept 2020 – Dec 2020**

- Analyzed data gained from particle size distribution tests to inform engineers how to deal with their bulk
- Designed different office layouts using PowerCADD under project constraints

Education & Awards

University of Waterloo | Honours Mechatronics Engineering **Sept 2020 - Apr 2026**

Lester B. Pearson C.I. Staff Scholarship Award **Jun 2020**